StudyPlanner

Full Stack Development - Project 1
Responsive Frontend Interface

DecodeLabs Industrial Training Kit | Batch 2026

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GitHub: github.com/sadaff-zahid/DecodeLabs-Internship

Live link: https://sadaff-zahid.github.io/DecodeLabs-Internship/

1. Project Overview

StudyPlanner is a simple web application that helps a student plan study tasks. It is built with plain HTML, CSS and basic JavaScript, with no frameworks, as the project brief requires. The focus of this milestone is the **responsive frontend interface**: a layout that adapts to phone, tablet and desktop.

Brief requirements and how they are met

Requirement	How StudyPlanner meets it
HTML, CSS and basic JavaScript	index.html, style.css, script.js. No frameworks
Responsive layout	Mobile-first; changes at 768px and 1024px
Clean, user-friendly UI	One clear form, simple task cards, big buttons
Semantic, accessible code	header, nav, main, aside, article, footer; labels; focus styles; skip link
2025 look	Mocha Mousse, Ethereal Blue, Moonlit Grey; Montserrat + Open Sans

2. Discovery: How Might We

Problem: Students forget deadlines and lose track of what to study. Many planner apps are too heavy or hard to use on a phone.

How Might We statement: *How might we help students quickly note down their study tasks and see their progress, on any device, with a very simple interface?*

Goals

- Add a task in under 10 seconds
- Easy to read and tap on a small screen
- See at a glance what is pending and what is done
- Work for everyone, including keyboard users

3. Research: Empathy Map

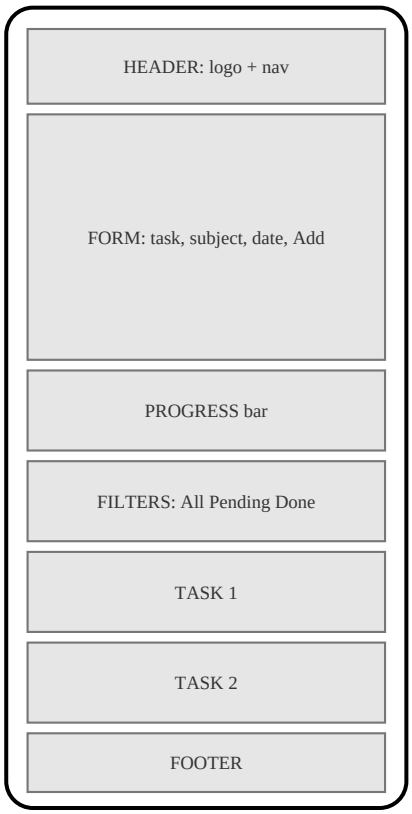
Persona: **A university student who studies between classes and mostly uses a phone.**

SAYS	**THINKS**
- "I always forget my deadlines." - "I want something quick, not complicated."	- "If I see my list, I will feel in control." - "Will this work on my phone?"
DOES	**FEELS**
- Checks the phone between classes - Writes tasks in chats or on paper - Studies close to deadlines	- Stressed when work piles up - Satisfied when a task is ticked off - Annoyed by tiny buttons

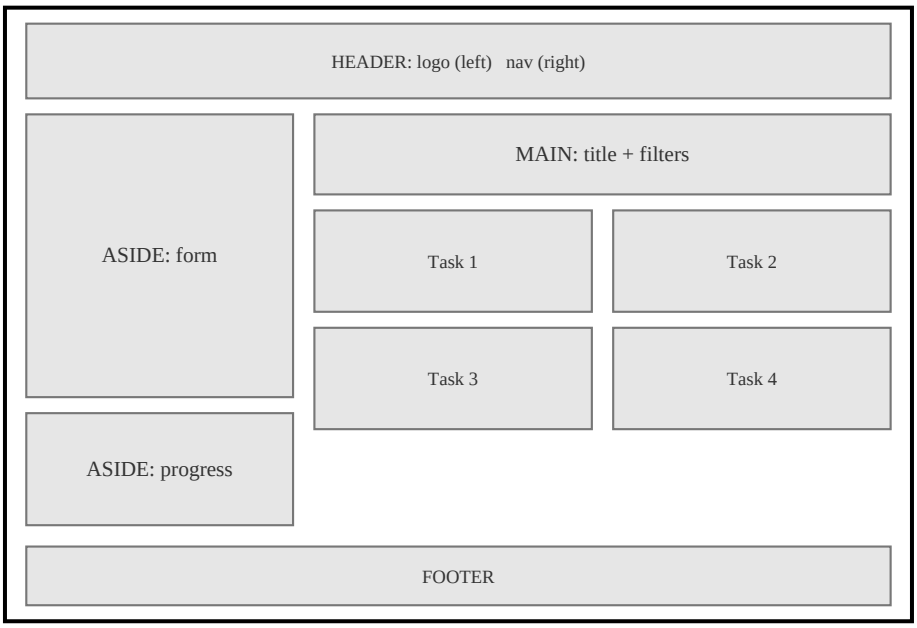
Pain points: forgetting deadlines, cluttered screens, tiny tap targets. **Answers in the design:** due date on every task, a single simple form, 44px buttons, and a progress bar.

4. Wireframes (grayscale, mobile-first)

Golden rule: if it does not work in grayscale, color will not fix it. The phone layout was planned first.



Mobile: one column



Desktop: aside + main (CSS Grid)

Layout plan

- Phone: single column. Header, form, progress, filters, tasks, footer
- Tablet (768px and up): sidebar on the left, tasks on the right
- Desktop (1024px and up): wider layout, tasks in two columns

5. Implementation

Semantic HTML5

- header with the logo, nav for links, main for the content

- aside holds the form and progress; article holds the task list; footer at the end

- Every input has a label; the task list uses aria-live so updates are announced

Layout and responsiveness

- CSS Grid for the page layout, Flexbox for header, buttons and cards

- Mobile-first: base styles are for phones, then min-width 768px and 1024px

- A container query changes each task card to a row when its box is wide

- clamp() makes headings scale smoothly

Visual design

Color	Role	Hex
Mocha Mousse	Stability	#A5856F (dark #6F4E3C for text/buttons)
Ethereal Blue	Trust	#A0D4E0
Moonlit Grey	Refinement	#F2F0EA

Fonts: Montserrat (headings) and Open Sans (text). Two families, three weights in total.

JavaScript logic

- Tasks are kept in a list (array); showTasks() redraws the page

- Add (with validation), Done/Undo, Delete, filter, progress bar

- Tasks are saved in the browser with localStorage

6. Testing and Audit

Lighthouse was run in Chrome on the live link (Mobile mode, 21 Oct 2026).

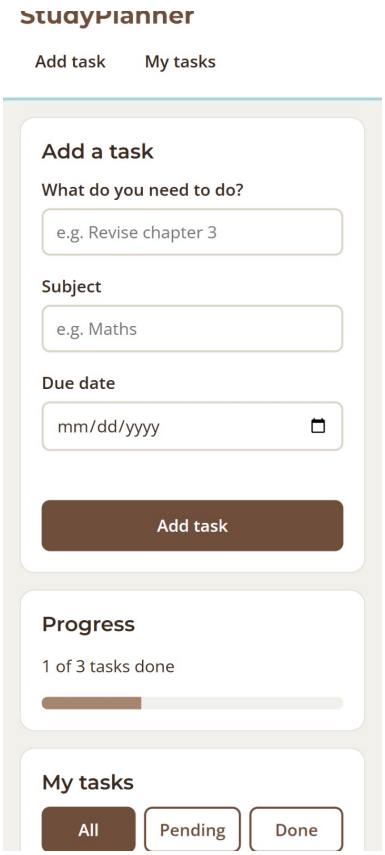
Check	How	Result
Performance	Lighthouse (Mobile)	71 / 100
Accessibility	Lighthouse	100 / 100
Best Practices	Lighthouse	100 / 100
SEO	Lighthouse	100 / 100
Phone view	Screenshot (section 8)	[x] Passed
Tablet view (768px)	Screenshot (section 8)	[x] Passed
Desktop view (1024px+)	Screenshot (section 8)	[x] Passed
Add / Done / Undo / Delete / Filter	Automated test of script.js (all steps passed)	[x] Passed
Keyboard and labels	Lighthouse Accessibility	[x] 100 / 100

Note on Performance (71): Lighthouse tests Mobile with a slow network. The page loads two Google Fonts, which delay the first paint. A future improvement is to load fewer font files or host them locally.

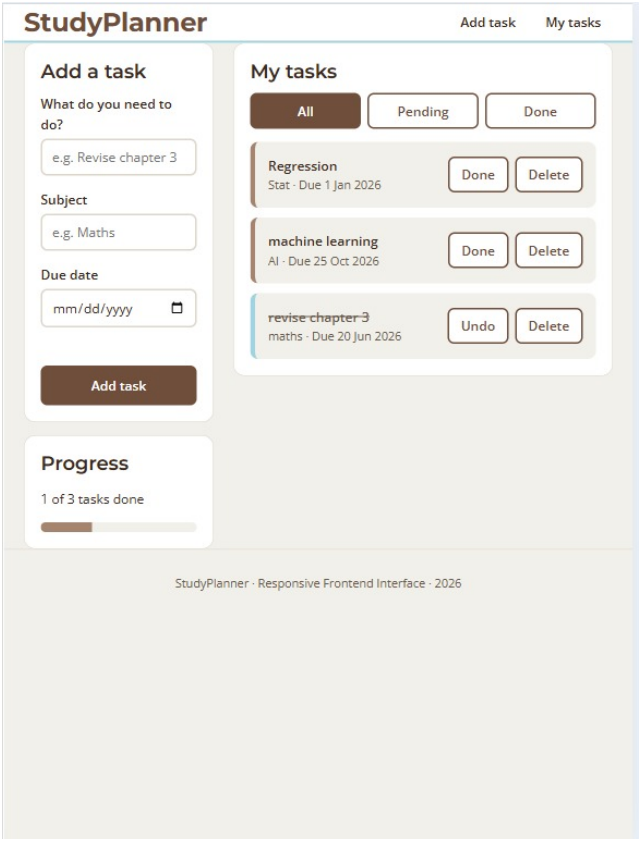
7. Roadmap Status

Step	Task	Status
1	Discovery (How Might We)	Done (section 2)
2	Wireframe, grayscale, mobile-first	Done (section 4)
3	Semantics (HTML landmarks)	Done (index.html)
4	Style (Grid/Flex, 2025 palette)	Done (style.css)
5	Logic (state, interactivity)	Done (script.js)
6	Audit (LCP/CLS, accessibility)	Done (section 6)

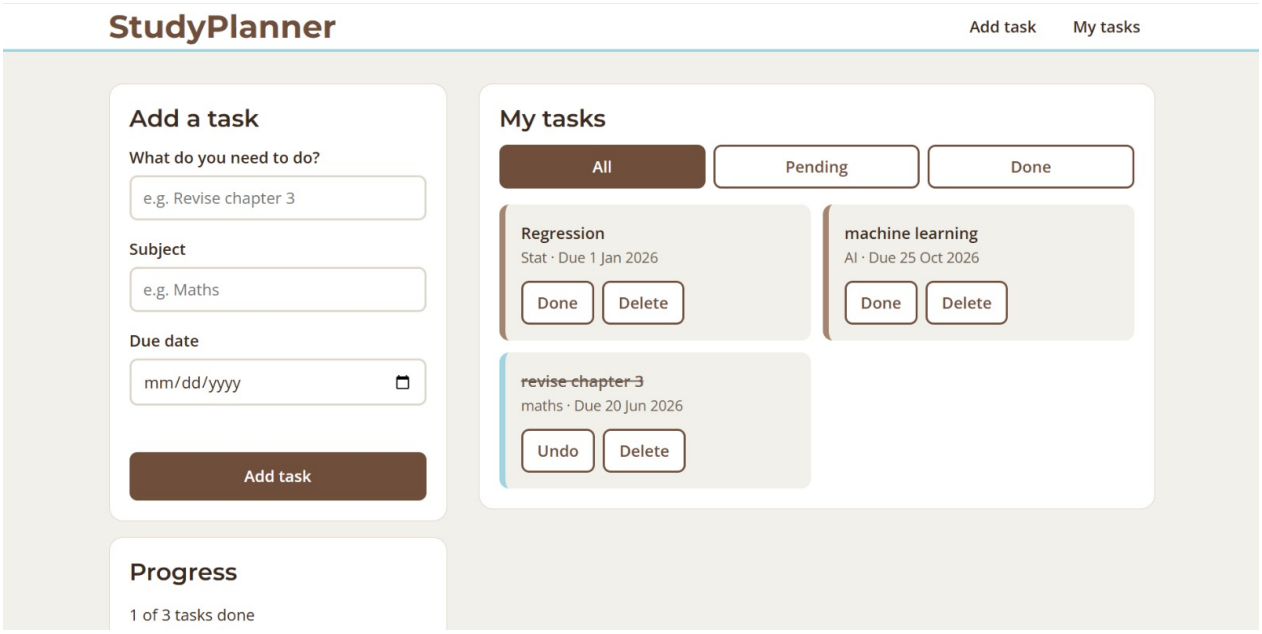
8. Screenshots



Phone (single column)



Tablet (sidebar + main)



Desktop (sidebar + tasks in two columns)

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